SMART RESUME BUILDER AND ANALYZER

A PROJECT REPORT

Submitted by

SUBIKSEN V S	**2303811710421160**
SUMAN S	**2303811710421161**
VISWA D	**2303811710421182**
VISWAK MANIMANNAN	**2303811710421183**

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(AUTONOMOUS)

SAMAYAPURAM–621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**SMART RESUME BUILDER AND ANALYZER**" has been carried out by the students **SUBIKSEN V S**, **SUMAN S, VISWA D, VISWAK MANIMANNAN.** The work reported here in is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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MR.M.A.PRASANNA M.E.,	**MR.R. RAJAVARMAN M.E.,(Ph.D.,)**
SUPERVISOR	**HEAD OF THE DEPARTMENT**
Assistant Professor,	Assistant Professor(Sr.Grade),
Department of CSE,	Department of CSE,
K Ramakrishnan College of Technology	K Ramakrishnan College of Technology
(Autonomous)	(Autonomous)
Samayapuram–621112	Samayapuram–621112

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Smart Resume Builder and Analyzer is a state-of-the-art, AI-powered career optimization platform designed to bridge the gap between job seekers and elite global recruiters. Built as a comprehensive executive suite, the system leverages local Large Language Models (Llama 3.2) via Ollama to provide high-precision, privacy-focused resume analysis without relying on external cloud APIs. The platform features an Elite Resume Builder that allows users to construct ATS-optimized, single-column professional resumes across 15 customizable sections, ensuring they meet the "top 0.001%" industry standards.A core innovation of the system is its Multi-Factor Audit Engine, which performs a rigorous 10-dimensional analysis of a candidate's profile, including Business ROI, Metrics-driven impact, and Professional Trajectory. Beyond static analysis, the platform includes an AI Interview Simulator that generates tailored behavioral and technical questions derived directly from the user's specific projects and achievements.By integrating deterministic AI scoring with a premium, high-performance user interface, the Smart Resume Builder and Analyzer serves as a unified career cockpit, empowering candidates to identify skill gaps, quantify their professional value, and secure positions at FAANG-tier organizations.

Keywords: Artificial Intelligence (AI), Natural Language Processing (NLP), Streamlit, Llama 3.2, ATS, Career Analytics,MySQL.

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SIGNATURE

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ATS	Application Tracking System
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
HTML	Hyper Text Markup Language
LLM	Large Language Model
NLP	Natural Language Processing
REST	Representational State Transfer
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Smart Resume Builder and Analyzer is a high-performance, intelligent career platform designed to empower job seekers in building world-class resumes and mastering competitive interview cycles. The system focuses on maximizing a candidate's visibility by providing deep, AI-driven profile audits along with integrated tools for document construction and market value estimation.

The platform allows users to create professional profiles, construct resumes from 15 elite sections, or upload existing documents for immediate assessment. Based on this data, the system utilizes local Llama 3.2 (LLM) technology to perform a 10-factor audit—analyzing everything from Business ROI and Metrics to Storytelling and Formatting. This helps users transform standard resumes into high-impact, FAANG-tier professional documents.

In addition to resume optimization, the project includes an AI Interview Prep module that generates sophisticated, tailored questions based on the candidate's specific achievements. Users can also access a Job Market Insight module to explore career paths and identify their demand level across top platforms like LinkedIn and Naukri. The system enables users to visualize their professional trajectory and identify critical skill gaps using interactive data metrics.

The system is designed with a premium, modular architecture consisting of six primary components: Authentication & Security, AI Resume Auditor, Elite Resume Builder, Interview Simulator, Job Market Insights, and the Executive Admin Dashboard. This ensures a seamless, high-performance experience from the initial profile setup to the final document generation.

Overall, the Smart Resume Builder and Analyzer acts as a complete "Career Cockpit," combining deterministic AI analysis, elite resume construction, and strategic interview preparation into a single, privacy-focused platform.

1.2 PROBLEM STATEMENT

In the current hyper-competitive global job market, both freshers and experienced professionals face significant challenges in securing high-tier employment, particularly at FAANG and Fortune 500 companies. One of the major issues is the lack of rigorous, factor-based resume analysis. Most candidates rely on generic AI tools or simple templates that do not evaluate critical professional dimensions such as Business ROI, metrics-driven impact, or executive storytelling. As a result, many resumes are rejected by sophisticated Applicant Tracking Systems (ATS) because they fail to quantify professional value effectively.

Another significant problem is the data privacy risk associated with cloud-based AI platforms. Professionals are often forced to upload sensitive personal information, including contact details and detailed career histories, to external servers just to receive basic feedback. This lack of localized, secure AI solutions prevents many security-conscious candidates from utilizing modern tools to optimize their profiles. Furthermore, many individuals struggle to understand the "seniority gap"—how their current experience translates into the next professional level—leading to misaligned job applications.

Existing platforms typically focus on only one aspect—either aesthetic design or keyword matching—but do not provide a "Career Cockpit" that integrates deep-factor auditing, elite document construction, and resume-based interview simulation. Without a centralized system that offers deterministic scoring and actionable improvement steps, users are left with "blind spots" in their professional presentation that only become apparent after a failed interview.

Therefore, there is a need for a unified, local-AI platform that provides deep-factor resume auditing, executive-level construction, and tailored interview simulation within a single, secure environment. Such a system can significantly improve a candidate's ATS visibility, narrative precision, and overall readiness for elite recruitment cycles. Furthermore, there is a limited emphasis on the "Impact-to-Metric" ratio in current builders. Existing tools rarely provide the analytical depth required to highlight leadership, cultural fit, and ROI, which are the primary factors that elite organizations use to differentiate top-tier talent from average candidates.

1.3 DOMAIN SPECIFICATION

The Smart Resume Builder and Analyzer project belongs to the domain of HR Technology and Privacy-Centric AI Career Systems, focusing on enhancing professional recruitment outcomes through advanced analytical and automated solutions. This domain integrates cutting-edge concepts from NLP, LLMs, and Full-stack Web Development to create a secure, high-performance ecosystem for modern job seekers.

The primary objective of this domain is to empower candidates with "Executive-Level" insights by bridging the gap between raw professional experience and the sophisticated requirements of modern ATS. It focuses on delivering tools that support high-fidelity resume construction, deterministic career auditing, and resume-derived interview simulation, all within a unified and privacy-focused local environment.

1.3.1 AI-Driven Career Auditing and Strategic Feedback Domain

This domain focuses on transforming static professional history into a dynamic, analytical report by auditing resumes across ten critical executive dimensions. The system utilizes local Large Language Models (Llama 3.2) to perform deep-factor analysis, providing quantitative scores (0-100) and qualitative feedback.

It includes multiple types of analytical feedback such as:

1. Business Value & ROI Analysis
2. Professional Trajectory Mapping
3. Metrics and Action Verb Auditing
4. Leadership and Cultural Fit Evaluation

This domain ensures that candidates receive an honest, recruiter-level assessment of their profile, identifying "Critical Red Flags" and providing actionable steps to improve their ATS visibility and executive presence.

1.3.2 Elite Document Engineering and ATS Optimization Domain

This domain deals with the structural engineering of top-tier professional documents. It enables users to construct resumes from the ground up using 15 customizable "Elite" sections designed to satisfy the most rigorous Applicant Tracking Systems.

The system acts as a bridge between:

- Raw Candidate Experience (Inputs)
- Recruiter-Ready PDF Output (Diamond Elite Templates)

It simplifies the process of formatting complex professional data, ensuring that font hierarchy, grid alignment, and content density meet the "top 0.001%" industry standards. This domain eliminates the need for external designers by providing automated, high-fidelity PDF generation that is both visually stunning and machine-readable.

1.3.3 Local Intelligence and Private Data Management Domain

This domain focuses on the secure management of sensitive professional data through a privacy-first, local-AI architecture. Instead of relying on vulnerable cloud APIs, the system manages all intelligence tasks locally using Ollama, ensuring that resumes and credentials never leave the user's machine.

It handles critical backend operations such as:

- Secure User Authentication and Credential Hashing
- Local Resume Ingestion and PDF Reading
- Encrypted Storage of Analysis Reports and Builder Data

This domain ensures that the user maintains complete sovereignty over their career data, providing a high-performance environment that combines the power of generative AI with the security of a localized database system.

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The aim of the Smart Resume Builder and Analyzer project is to develop a comprehensive, privacy-focused executive platform that empowers job seekers to optimize their professional profiles through deep-factor AI analysis and high-fidelity document engineering. The system is designed to bridge the gap between individual career history and the rigorous standards of elite global recruiters by offering a unified solution for resume auditing, premium document construction, and personalized interview preparation.

1.4.2 Objective

The main objectives of the Smart Resume Builder And Analyzer project are:

- To develop a 10-Factor AI Auditor for providing rigorous, multi-dimensional scores and actionable feedback on resume quality.
- To implement a Resume-Derived Interview Simulator that generates sophisticated behavioral and technical questions based on user projects.
- To provide an integrated Market Insight Module for role matching, salary estimation, and demand analysis across top job platforms.
- To ensure Data Sovereignty by integrating a local LLM (Ollama) that processes all sensitive information on the user's local machine.
- To improve Candidate Confidence by identifying "Critical Red Flags" and providing a clear, step-by-step action plan for professional improvement.
- To integrate future-ready features like AI Skill Gap Prediction and automated tailoring of resumes to specific job descriptions.

1.5 SCOPE OF THE PROJECT

The scope of the Smart Resume Builder and Analyzer project focuses on developing a high-performance, privacy-centric platform that supports job seekers in their career optimization by integrating 10-factor resume auditing, elite document construction, and tailored interview simulation into a single executive dashboard.

The project primarily targets final-year and pre-final-year students from both IT and core engineering branches such as Computer Science, Electrical, Mechanical, and Civil Engineering. It provides a structured environment where students can improve their interview skills based on their individual profiles and career goals.

In addition to resume auditing, the system includes a Job Market Insight module that allows users to determine their market value and discover relevant roles across platforms like LinkedIn and Naukri. The integrated Elite Resume Builder enables the generation of top-tier, ATS-optimized PDFs, ensuring that the user's professional presentation is consistent across all application channels.

CHAPTER 2

LITERATURE SURVEY

2.1 Automated Resume Classification Using Machine Learning

The research work titled *"Automated Resume Classification Using Machine Learning"* focuses on improving the recruitment process by automating the classification of resumes into different job categories. Traditional resume screening is time-consuming and prone to human bias, which makes automation essential. By applying concepts from Machine Learning, the system is designed to efficiently handle large volumes of resumes and categorize them based on predefined domains such as IT, management, or healthcare.

The system primarily relies on Natural Language Processing techniques to process and analyze resume content. It involves steps such as tokenization, stop-word removal, and feature extraction to convert unstructured text into a structured format.

Various classification algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Decision Trees are employed in the study to categorize resumes accurately. Among these, SVM often provides better accuracy due to its ability to handle high-dimensional data. The model is trained on a dataset of labeled resumes, allowing it to predict the category of new resumes with reasonable precision.

The system also evaluates performance using metrics like accuracy, precision, recall, and F1-score. These evaluation techniques ensure that the model is reliable and effective in real-world scenarios. However, the study highlights certain challenges, such as handling ambiguous terms, lack of standard resume formats, and difficulty in interpreting contextual meaning.

In conclusion, the paper demonstrates that machine learning-based resume classification significantly enhances recruitment efficiency by automating the screening process. While the system improves speed and accuracy, future improvements can include deep learning models and better contextual understanding to further refine classification and provide more intelligent insights.

2.2 Resume Parsing and Semantic Analysis Using NLP Techniques

The research work titled *"Resume Parsing and Semantic Analysis Using NLP Techniques"* focuses on extracting structured information from unstructured resume documents. In traditional systems, resumes are manually reviewed, which is inefficient and error-prone. This study introduces automated parsing using Natural Language Processing to convert raw resume text into meaningful and machine-readable data, improving the efficiency of recruitment systems.

The system processes resumes through several NLP stages such as tokenization, stemming, and part-of-speech tagging. It also uses Named Entity Recognition (NER) to identify key elements like candidate name, skills, education, and work experience. These techniques help in organizing resume data into predefined categories, making it easier for systems to analyze and compare candidate profiles.

A key contribution of this approach is semantic analysis, which goes beyond keyword matching. Instead of simply detecting words, the system attempts to understand the meaning and context behind them. For example, it can recognize that "software developer" and "programmer" are closely related roles. This improves the accuracy of candidate evaluation compared to traditional keyword-based systems.

The system is often integrated with databases and recruitment platforms to enable efficient searching and filtering of candidates. By structuring resume data, it allows recruiters to quickly identify suitable candidates based on specific criteria. However, challenges such as variations in resume formats, use of abbreviations, and inconsistent terminology can affect parsing accuracy.

In conclusion, this study highlights the importance of NLP-based resume parsing and semantic analysis in modern recruitment systems. While it significantly improves data extraction and understanding, further advancements in contextual analysis and domain-specific language models are required to enhance accuracy and adaptability in real-world applications.

2.3 Intelligent Resume Analyzer for Job Recommendation Systems

The research work titled *"Intelligent Resume Analyzer for Job Recommendation Systems"* focuses on enhancing recruitment by matching candidate resumes with suitable job roles. Traditional job matching systems often rely on basic keyword searches, which may not accurately reflect a candidate's capabilities. This study introduces an intelligent system that leverages Machine Learning techniques to provide more accurate and personalized job recommendations.

The system utilizes Natural Language Processing to extract important information such as skills, qualifications, and experience from resumes. It then converts both resumes and job descriptions into vector representations using techniques like TF-IDF (Term Frequency–Inverse Document Frequency). These vectors help in comparing the similarity between resumes and job requirements in a mathematical manner.

A key component of the system is the use of similarity measures, particularly cosine similarity, to determine how closely a resume matches a job description. Based on the similarity score, the system ranks job opportunities for each candidate. This approach improves the efficiency of job recommendation by ensuring that candidates are matched with roles that closely align with their profiles.

The system also incorporates feedback mechanisms where users can refine their profiles based on recommendations. This helps in improving future matching accuracy. However, the study identifies certain limitations, such as difficulty in understanding contextual meaning, handling synonyms effectively, and adapting to domain-specific requirements.

In conclusion, the intelligent resume analyzer significantly improves job recommendation systems by combining NLP and machine learning techniques. While it enhances matching accuracy and user experience, future work can focus on deep learning models and semantic understanding to provide more precise and context-aware recommendations.

2.4 ATS-Based Resume Evaluation System Using Text Mining

The research work titled *"ATS-Based Resume Evaluation System Using Text Mining"* focuses on automating the screening process of resumes using Applicant Tracking Systems (ATS). In modern recruitment, organizations receive a large number of applications, making manual evaluation inefficient. This study highlights how techniques from Text Mining are used to filter and rank resumes efficiently based on predefined criteria.

The system analyzes resumes by extracting relevant keywords and comparing them with job descriptions. It relies on Natural Language Processing techniques such as tokenization and keyword extraction to identify important terms like skills, qualifications, and experience. These keywords play a crucial role in determining whether a resume passes the initial screening stage.

A scoring mechanism is implemented to rank resumes based on keyword relevance, formatting, and content structure. Resumes that closely match job requirements receive higher scores, increasing their chances of selection. This approach helps recruiters quickly shortlist suitable candidates without manually reviewing each application.

However, the system has certain limitations. It heavily depends on keyword matching, which may lead to the rejection of qualified candidates who use different terminology. Additionally, it lacks deep semantic understanding and may not accurately interpret the context of skills or experience mentioned in the resume.

In conclusion, the ATS-based resume evaluation system improves recruitment efficiency by automating resume screening and ranking. While it is effective in handling large volumes of applications, future enhancements should focus on incorporating semantic analysis and contextual understanding to overcome the limitations of keyword-based evaluation systems.

2.5 AI-Powered Resume Builder and Feedback System

The research work titled *"AI-Powered Resume Builder and Feedback System"* focuses on developing an intelligent platform that assists users in both creating and improving resumes. Unlike traditional tools, this system integrates features of resume building and analysis using Artificial Intelligence to provide real-time suggestions. It aims to enhance resume quality by guiding users in structuring content, selecting appropriate keywords, and maintaining professional formatting.

The system uses Natural Language Processing techniques to analyze the textual content of resumes. It evaluates grammar, sentence structure, and clarity to ensure the resume meets professional standards. Additionally, it identifies missing sections, weak phrases, and irrelevant information, helping users refine their resumes effectively.

A key feature of this system is its feedback mechanism, which provides suggestions such as skill enhancements, keyword optimization, and formatting improvements. It may also generate an ATS score to indicate how well the resume performs in automated screening systems. This enables users to understand their resume's strengths and weaknesses in a measurable way.

The system can also recommend relevant skills and job roles based on the user's profile. By analyzing industry trends and job descriptions, it helps users align their resumes with market demands. However, challenges include maintaining accuracy in recommendations and adapting to different domains and job roles.

In conclusion, the AI-powered resume builder and feedback system represents a significant advancement in resume preparation tools. It combines automation, intelligent analysis, and personalized feedback to improve user outcomes. Future developments can focus on deeper contextual understanding, domain-specific intelligence, and integration with real-time job market data for more effective and adaptive resume solutions.

2.6 Deep Learning-Based Resume Screening and Ranking System

The research work titled *"Deep Learning-Based Resume Screening and Ranking System"* focuses on improving the accuracy of candidate selection using advanced models from Deep Learning. Traditional machine learning approaches rely heavily on manual feature extraction, whereas deep learning models automatically learn complex patterns from large datasets. This system aims to enhance recruitment by providing more precise ranking and screening of resumes.

The system utilizes techniques from Natural Language Processing along with neural networks such as Recurrent Neural Networks (RNN) and Transformer-based models. These models process resume text and job descriptions to capture contextual meaning, making the system more effective than keyword-based or shallow learning approaches.

A key feature of this approach is semantic matching, where the system understands the relationship between skills, experience, and job requirements. Instead of relying only on exact keyword matches, it identifies similar concepts and evaluates candidate suitability more accurately. This leads to better ranking of resumes based on relevance and quality.

The model is trained on large datasets of resumes and job descriptions, improving its ability to generalize across different domains. It also uses ranking algorithms to prioritize candidates, helping recruiters make faster and more informed decisions. However, the system requires significant computational resources and large training data, which can be a limitation.

In conclusion, deep learning-based resume screening systems provide a more advanced and accurate solution compared to traditional methods. By incorporating contextual understanding and semantic analysis, these systems significantly improve candidate evaluation. Future work can focus on reducing computational cost and enhancing explainability to make these systems more practical and transparent.

2.7 Cloud-Based Resume Builder and Recruitment Management System

The research work titled *"Cloud-Based Resume Builder and Recruitment Management System"* focuses on providing scalable and accessible resume services through cloud technology. Traditional resume tools are often limited to local systems, but this approach leverages Cloud Computing to enable users to create, store, and manage resumes online from anywhere. It improves accessibility, collaboration, and data management in the recruitment process.

The system allows users to build resumes using web-based interfaces with dynamic templates and real-time editing features. It integrates databases to store user profiles, resumes, and job-related data securely. Cloud platforms ensure data availability and backup, reducing the risk of data loss and enabling seamless updates across devices.

A key feature of this system is integration with recruitment modules such as job posting, application tracking, and candidate management. Recruiters can access resumes directly from the cloud and filter candidates based on specific requirements. This centralized system enhances communication between job seekers and employers.

The platform may also incorporate Artificial Intelligence features such as resume analysis, keyword suggestions, and automated feedback. These capabilities help users optimize their resumes for better visibility in recruitment systems. However, concerns such as data privacy, security risks, and dependency on internet connectivity remain challenges.

In conclusion, cloud-based resume builder and recruitment systems provide a flexible and efficient solution for modern hiring needs. They enhance accessibility, scalability, and collaboration while integrating intelligent features. Future improvements can focus on strengthening security measures and incorporating advanced analytics for better decision-making in recruitment processes.

2.8 Online Resume Recommendation System Using Collaborative Filtering

The research work titled *"Online Resume Recommendation System Using Collaborative Filtering"* focuses on improving job matching by suggesting suitable opportunities based on user behavior and preferences. Traditional systems rely mainly on keyword matching, but this approach introduces recommendation techniques from Recommender Systems to provide more personalized results for job seekers.

The system uses collaborative filtering methods to analyze patterns among users with similar profiles, skills, and job interests. By comparing user interactions such as applied jobs, searched roles, and viewed listings, it predicts and recommends relevant job opportunities. This helps in providing tailored suggestions rather than generic job listings.

In addition, the system may combine Machine Learning techniques to improve recommendation accuracy. Hybrid models integrate both collaborative and content-based filtering, where resume content is analyzed alongside user behavior. This enhances the quality of recommendations by considering both user preferences and job requirements.

A key advantage of this system is its ability to continuously improve over time as more user data becomes available. It adapts to changing trends and user interests, making job recommendations more relevant. However, challenges such as cold start problems (new users with no data) and data sparsity can affect performance.

In conclusion, collaborative filtering-based resume recommendation systems offer a personalized and dynamic approach to job matching. By leveraging user behavior and machine learning techniques, these systems enhance the recruitment experience. Future work can focus on overcoming cold start issues and integrating deeper semantic understanding for more accurate and context-aware recommendations.

2.9 Hybrid Resume Analysis System Using NLP and Machine Learning

The research work titled *"Hybrid Resume Analysis System Using NLP and Machine Learning"* focuses on combining multiple intelligent techniques to improve resume evaluation. Traditional systems either rely only on keyword matching or basic classification, but this hybrid approach integrates methods from Natural Language Processing and Machine Learning to achieve better accuracy and deeper analysis.

The system first processes resumes using NLP techniques such as tokenization, lemmatization, and entity extraction to identify key components like skills, education, and experience. This structured data is then fed into machine learning models to classify and evaluate resumes based on job requirements. The combination of these techniques helps in capturing both syntactic and semantic information.

A major feature of the hybrid system is its ability to perform both classification and scoring. It not only categorizes resumes into domains but also assigns scores based on relevance, completeness, and quality. This dual functionality makes it more effective compared to standalone systems that perform only one task.

The system also supports job-resume matching by comparing extracted features with job descriptions. It improves matching accuracy by considering contextual relationships between terms rather than relying solely on exact keyword matches. However, challenges such as handling diverse resume formats and domain-specific terminology still exist.

In conclusion, the hybrid resume analysis system provides a balanced and efficient solution by combining NLP and machine learning techniques. It enhances classification, scoring, and matching processes, making it highly useful in recruitment systems. Future improvements can focus on integrating deep learning models and real-time feedback mechanisms for even more advanced and adaptive resume analysis.

2.10 Resume Screening System Using Ontology-Based Semantic Analysis

The research work titled *"Resume Screening System Using Ontology-Based Semantic Analysis"* focuses on improving resume evaluation by incorporating domain knowledge into the analysis process. Traditional systems rely heavily on keyword matching, which often fails to capture the true meaning of candidate skills. This study introduces the use of Ontology to represent knowledge in a structured and meaningful way for better resume understanding.

The system builds a domain-specific ontology that defines relationships between skills, job roles, qualifications, and experience. For example, it can understand that "Java" is related to "object-oriented programming" and "software development." This allows the system to interpret resumes more intelligently compared to basic text-matching approaches.

It integrates Natural Language Processing techniques to extract key information from resumes and map them to the ontology structure. By doing so, the system performs semantic matching between resumes and job descriptions, improving the accuracy of candidate selection even when different terminology is used.

A key advantage of this approach is its ability to reduce ambiguity and improve contextual understanding. It enhances decision-making by considering relationships between concepts rather than isolated keywords. However, building and maintaining domain-specific ontologies is time-consuming and requires expert knowledge, which can be a limitation.

In conclusion, ontology-based resume screening systems provide a more intelligent and semantically rich approach to candidate evaluation. They overcome many limitations of keyword-based systems and improve matching accuracy. Future work can focus on automating ontology creation and integrating it with advanced AI models for more scalable and efficient resume analysis systems.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

In the current professional landscape, job seekers and career professionals rely on multiple independent platforms and fragmented methods for resume optimization and interview preparation. There is no unified, local system that integrates all essential executive functionalities—such as 10-factor resume auditing, premium document construction, and resume-based interview simulation—into a single secure environment.

For resume analysis, users commonly depend on cloud-based AI tools or basic online ATS checkers. These resources present two major flaws: they require the uploading of highly sensitive personal data to external servers, creating significant privacy risks, and they often provide generalized, vague feedback. Most existing systems do not evaluate a resume's professional trajectory, Business ROI, or narrative depth, leading to resumes that may contain the right keywords but fail to demonstrate a candidate's actual executive value.

Similarly, interview preparation is typically handled through generic question banks or standard behavioral guides. These resources do not consider the user's individual projects, specific technical achievements, or the unique context of their career history. As a result, candidates often enter interviews with a "generic" preparation strategy that fails to highlight their specific strengths or address the critical red flags identified in their unique resume.

Another major limitation in the current scenario is the lack of Localized Intelligence and Security. Most modern A.Moreover, there is no Integrated Career Narrative. A user may design a resume on one platform, check its score on another, and practice questions on a third. This disconnected approach often results in a mismatch between the professional brand presented on paper and the candidate's performance during the interview. Without a centralized "Career Cockpit" that tracks the evolution of the resume alongside interview readiness, professionals struggle to maintain a cohesive and competitive professional identity.

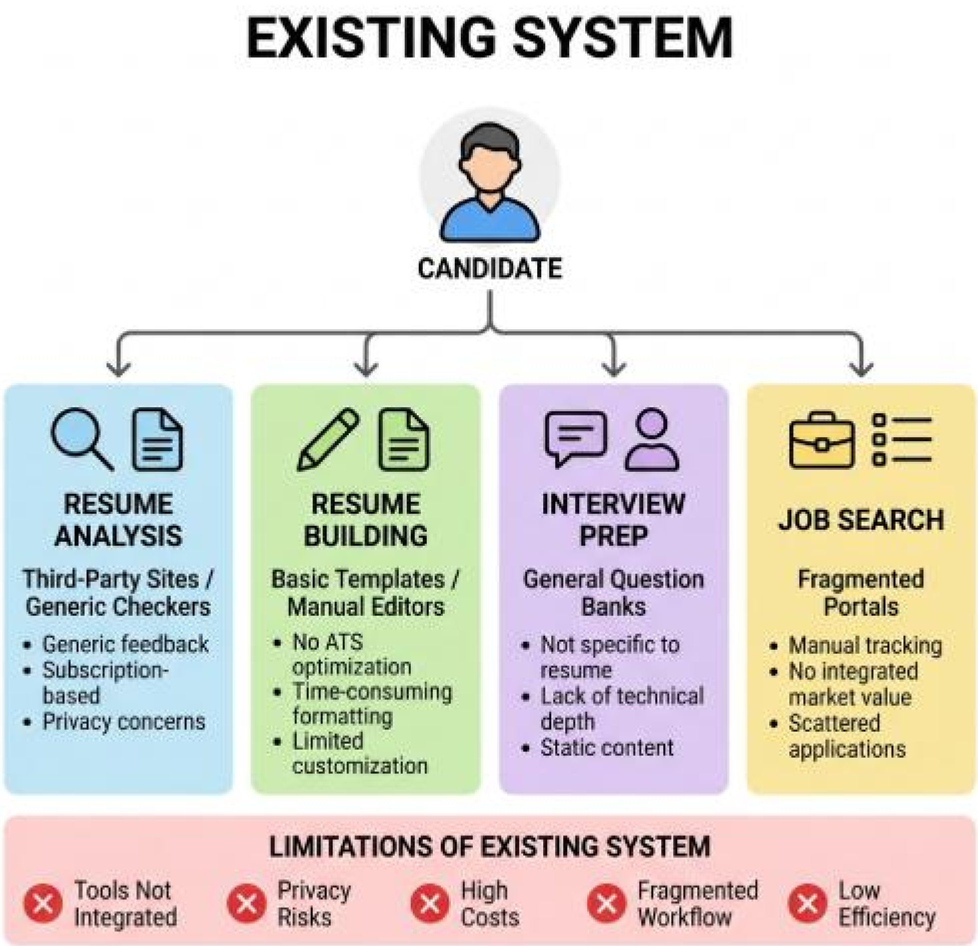

Figure 3.1 Existing System

3.2 DISADVANTAGES

- Fragmented across multiple platforms.
- High privacy and security risks.
- Vague and non-actionable feedback.
- Non-personalized interview preparation.
- Manual and tedious formatting.
- Expensive recurring subscription fees.
- Lack of integrated application tracking.
- Complete dependence on cloud APIs.

CHAPTER 4

PROBLEMS IDENTIFIED

Based on the analysis of the existing system, several key problems have been identified that affect a candidate's ability to optimize their career trajectory and prepare for elite recruitment cycles.

One of the major problems is the lack of deterministic, factor-based feedback. Most available platforms provide generic scores or simple keyword matches that are not tailored to a candidate's specific executive value, metrics, or ROI. This leads to ineffective resume optimization where the document looks good but fails to survive rigorous ATS and recruiter screening.

Candidates also face severe privacy and data security vulnerabilities. Since most modern AI tools are cloud-based, users must upload sensitive career data to external servers. This lack of a localized, secure environment creates a risk of data harvesting and prevents professionals from having full sovereignty over their professional history.

In addition, the process of resume construction is often manual and tedious. Traditional builders lack intelligent design systems that automatically optimize layout, font hierarchy, and content density for ATS compliance, forcing users to spend hours on repetitive formatting tasks rather than focusing on their professional narrative.

Another significant problem is the disjointed nature of interview preparation. Current prep tools do not analyze a user’s unique resume projects to generate tailored behavioral or technical questions. This results in a mismatch between the achievements listed on the resume and the candidate's ability to articulate them during the interview.

Finally, there is a lack of clear performance evaluation and red-flag identification. Existing systems rarely identify "Critical Red Flags" or provide a step-by-step action plan, preventing users from understanding their specific weaknesses and improving their professional presentation effectively..

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system, Smart Resume Builder and Analyzer, is a high-performance, privacy-centric platform designed to assist professionals and students in maximizing their career potential through an integrated "Career Cockpit." It provides a unified solution that combines deep-factor resume auditing, elite document construction, personalized interview preparation, and job market matching into a single secure interface.

In this system, users can either build a premium resume from 15 customizable elite sections or upload an existing PDF for a 10-factor audit. Based on this input, the platform utilizes a local Llama 3.2 model to analyze the document and provide deterministic scores on dimensions like Business ROI, Professional Trajectory, and Metrics

The system also includes an AI Interview Prep module that generates sophisticated behavioral and technical questions derived directly from the candidate's unique projects. Additionally, the Job Market Insight module helps users identify their market value, salary range, and specific role matches on platforms like LinkedIn and Naukri, providing a strategic edge in the job search process.

Security and data privacy are the foundation of the system design. By leveraging Ollama for local inference and MySQL for secure data storage, all user credentials, resumes, and analysis reports are kept strictly confidential. The system employs modern hashing and authentication mechanisms to ensure that the integrity of professional data is never compromised.

Overall, the Smart Resume Builder and Analyzer evolves from a simple preparation tool into a comprehensive career support ecosystem. It successfully bridges the gap between raw professional experience and the elite standards of global hiring markets, empowering users to own their career narrative with confidence and precision.

5.2 PROPOSED SYSTEM ARCHITECTURE

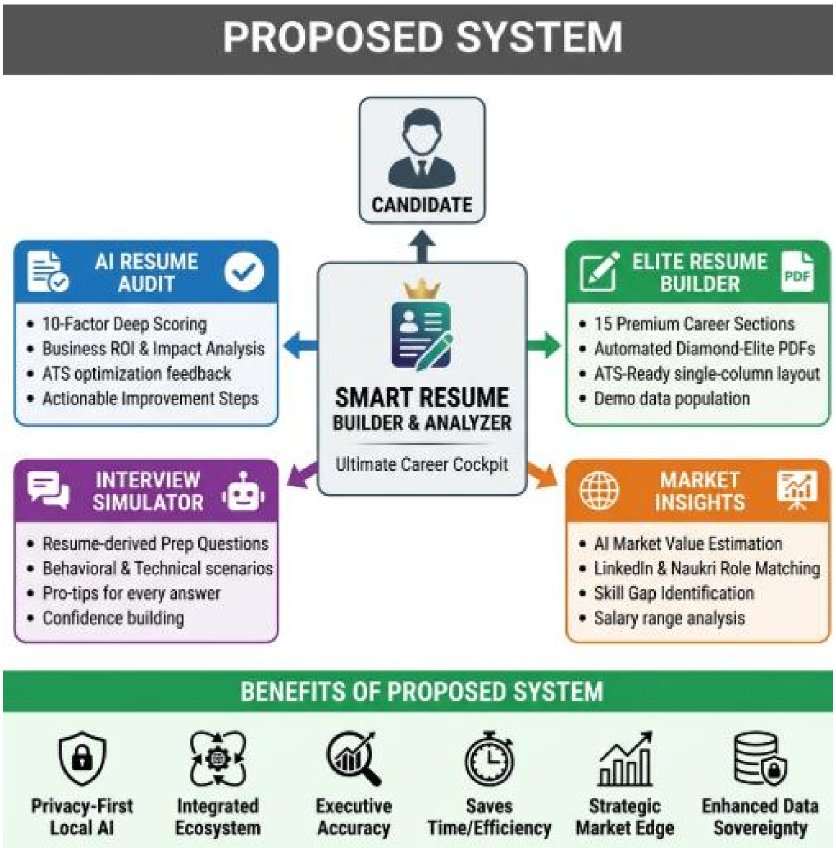

Figure 4.1 Proposed System

5.3 ADVANTAGES

- Unified all-in-one career platform.
- Privacy-focused local AI architecture.
- Rigorous 10-factor executive auditing.
- ATS-optimized elite PDF generation.
- Personalized, resume-based interview prep.
- Real-time market value estimation.
- Consistent, deterministic AI scoring.
- Robust self-healing data processing.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

The hardware requirements for developing and running the Smart Resume Builder And Analyzer are minimal, as it is a web-based application. The system can operate efficiently on standard computing devices.

- Processor : Intel i3 or higher
- RAM : Minimum 4 GB (8 GB recommended)
- Storage : At least 20 GB free space
- System Type : 64-bit system
- Device : Laptop / Desktop / Smartphone
- Internet Connection : Stable internet required

In addition to the basic hardware requirements, the performance and user experience of the Smart Resume Builder And Analyzer can be further enhanced with moderate upgrades. On the development side, a system with a multi-core processor such as Intel i5 or higher, along with 8 GB or more RAM, can significantly improve the efficiency of tasks like running local servers, database management, and testing machine learning models for resume analysis.

For deployment, cloud-based infrastructure can be utilized to ensure scalability, reliability, and high availability, allowing the system to handle multiple users simultaneously without performance degradation.

From the user perspective, devices with updated browsers and modern hardware configurations can provide a smoother and more responsive interface, especially when interacting with dynamic features like dashboards or real-time feedback systems. Additionally, implementing content delivery networks (CDNs) can reduce latency and improve page loading speed by delivering content from servers closest to the user's location.

Overall, these enhancements strengthen the system's performance, scalability, and reliability, making Smart Resume Builder And Analyzer more efficient and capable of supporting a growing number of users.

6.2 SOFTWARE REQUIREMENTS

The software requirements include tools and technologies used for development and execution of the Smart Resume Builder And Analyzer.

Operating System:

- Windows 10 or above
- Linux / macOS (optional)

Frontend Technologies:

- Streamlit
- HTML
- CSS
- JavaScript

Backend Technologies:

- Python

Database:

- MySQL

Development Tools:

- Visual Studio Code
- Git & GitHub (for version control)

AI & Processing:

- NLP libraries for resume parsing
- LLMa 3.2

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The Smart Resume Builder and Analyzer system is divided into six major modules, each responsible for handling a specific executive functionality within the platform. These modules work together to provide a complete career optimization and data-driven auditing system.

The system consists of the following six main modules:

- Authentication & Security Module
- Executive Resume Auditor Module
- Elite Resume Builder Module
- AI Interview Simulator Module
- Job Market Matchmaker Module
- Executive Admin Dashboard

7.2 MODULE DESCRIPTION

7.2.1 Authentication & Security Module

This module serves as the primary gateway to the platform, managing user identity and data sovereignty through a highly secure registration and login framework that utilizes advanced hashing algorithms for credential protection. It is designed to ensure a private and individualized experience, allowing users to save their career progress, builder data, and analysis history within a localized database without risk of exposure to external servers. Beyond basic access, the module handles session persistence and data integrity, ensuring that all professional documents, contact essentials, and executive reports remain confidential and accessible only to the account owner, thereby establishing a foundation of trust and reliability for professionals handling sensitive career information.

7.2.2 Executive Resume Auditor Module

As the central intelligence engine of the system, this module transforms raw resume data into a strategic analytical report by performing a rigorous 10-factor audit using a locally hosted Llama 3.2 Large Language Model. The system ingests PDF documents, parses complex professional metadata, and evaluates the content across critical dimensions such as Business Value ROI, Professional Trajectory, and Metrics-driven impact to provide a deterministic ATS score between 0 and 100. For every section analyzed, the module generates a multi-factor report including a 1-sentence verdict, structured strengths and weaknesses, and a list of "Critical Red Flags," providing users with a clear, data-backed roadmap to achieving elite-tier professional visibility and recruiter compliance.

7.2.3 Elite Resume Builder Module:

This module is dedicated to the high-fidelity engineering of professional resumes, offering a comprehensive suite of 15 customizable sections that allow users to detail their career objective, technical arsenal, innovation portfolio, and distinguished awards with precision. It features a sophisticated "Diamond-Elite" PDF generation engine that automatically applies executive-level formatting, including pixel-perfect grid alignment and high-contrast typography, to create a single-column document optimized for modern Applicant Tracking Systems. To assist users in understanding high-tier documentation standards, the module includes an integrated "Demo Mode" that instantly populates the builder with expert-level sample content, enabling a faster and more efficient construction process while maintaining the highest standards of professional branding and visual hierarchy.

7.2.4 AI Interview Simulator Module:

The Interview Simulator module provides a sophisticated preparation environment by generating personalized, high-stakes interview questions that are derived directly from the projects, skills, and leadership roles identified in the user's analyzed resume. By balancing behavioral scenarios with deep technical inquiry and system-design questions, the module replicates the intensity of elite recruitment cycles at top-tier technology firms like Google or Meta. Each generated question is accompanied by a strategic "Pro-Tip" that guides the candidate on how to structure their response effectively, ensuring their verbal articulation is as polished and impact-oriented as their written profile, thereby bridging the critical gap between getting an interview and securing a job offer.

7.2.5 Job Market Matchmaker Module:

This module functions as an intelligent career guide by analyzing a candidate's analyzed skill set against real-world hiring trends to provide real-time market value estimations and salary-range predictions. It simplifies the job discovery process by suggesting the most compatible career paths and generating tailored search queries for major global job portals such as LinkedIn and Naukri, ensuring that users target roles where they have the highest probability of success. Furthermore, the module identifies critical "Skill Gaps" by comparing the user's current profile against the requirements of their target roles, providing a strategic edge by highlighting the exact technologies or competencies the user needs to acquire to maximize their marketability and demand level in the competitive job market.

7.2.6 Executive Admin Dashboard

The Admin Dashboard is a robust management and visualization hub that provides administrators and institutional career centers with a centralized overview of all resume audits, builder submissions, and user progress metrics managed within the platform. It utilizes interactive tables and data-centric displays to track talent distribution across different candidate levels, recommended roles, and average ATS scores, making it an invaluable tool for organizations aiming to monitor placement readiness and talent quality. By offering deep insights into the collective performance of the user base, the dashboard allows for the identification of broader talent trends and institutional skill gaps, facilitating a more informed and data-driven approach to career counseling and professional development within an organization.

CHAPTER 8
SYSTEM TESTING

8.1 UNIT TESTING

Unit testing involves testing individual components or logic blocks of the system independently to ensure that each unit performs as expected before being integrated into the larger workflow. In the Smart Resume Builder and Analyzer, unit testing is performed on:

- **User Authentication:** Testing login/signup with hashed credential verification.
- **PDF Data Extraction:** Verifying that the pdf_reader correctly extracts text from various resume formats.
- **AI Prompt Logic:** Ensuring that the prompt generator correctly injects resume data into the Llama-3 instruction set.
- **JSON Self-Healing Utility:** Testing the robust_json_load function with malformed AI responses to ensure data recovery. Each function is tested with edge-case inputs to identify and fix errors early, ensuring high reliability at the component level.

8.2 INTEGRATION TESTING

Integration testing is performed to ensure that different modules of the system work together correctly, particularly focusing on the flow of data between the AI engine and the user interface. In this system, integration testing includes:

- **AI-to-Dashboard Sync:** Verifying that nested AI JSON data is correctly flattened and rendered into the 10 horizontal navigation tabs.
- **Builder-to-PDF Pipeline:** Ensuring that the data collected in the 15-section builder is correctly passed to the FPDF engine for document generation.
- **Database-to-Admin Feed:** Checking the real-time update of the executive dashboard when a new analysis is submitted. This testing ensures smooth communication between the local LLM, the MySQL database, and the Streamlit (communication between the frontend and the backend).

8.3 SYSTEM TESTING

System testing involves testing the complete platform as a single cohesive unit to ensure it meets all high-level functional requirements. In the Smart Resume Builder and Analyzer, system testing verifies:

- **End-to-End Workflow:** Testing the full journey from account creation to generating a Diamond-Elite PDF and receiving a 10-factor audit report.
- **Global Styling Integrity:** Ensuring that the Premium Dark Theme and custom CSS are consistently applied across all modules.
- **Resource Management:** Verifying that the application correctly handles PDF storage and database commits during high-load operations. This phase ensures that the system behaves as expected in real-world professional scenarios.

8.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate the system's responsiveness and stability, particularly regarding the execution of local AI models. In this system, performance testing involves:

Evaluating the speed of generating high-fidelity PDF documents with complex grid layouts and testing the responsiveness of the Admin Dashboard when handling hundreds of professional submissions are tested. This testing ensures that the local hardware can handle the computational load of the AI model without system crashes or excessive delays.

8.5 USABILITY TESTING

Usability testing focuses on evaluating the user experience, accessibility, and intuitive nature of the executive interface. In this system, usability testing involves:

Navigation Fluidity: Checking the ease of switching between the Builder, Auditor, and Interview Prep tabs.

Aesthetic Clarity: Ensuring that high-contrast text and labels are visible against the dark cinematic background.

CHAPTER 9
RESULT AND DISCUSSION

- The Smart Resume Builder and Analyzer was successfully developed and tested to provide an integrated, privacy-focused platform for professional career optimization and elite document engineering. The system effectively combines multi-factor resume auditing, premium PDF construction, and personalized interview preparation into a single high-performance web application.
- The implementation of the system produced the following key results:

  1. **Successful Local AI Integration:** The platform successfully integrated Llama 3.2 via Ollama, enabling complex career analysis to be performed locally without cloud dependencies.
  2. **Deterministic 10-Factor Auditing:** The system accurately parses uploaded resumes and provides rigorous scores and actionable feedback across ten critical professional dimensions.
  3. **High-Fidelity Document Generation:** The Elite Resume Builder successfully generates ATS-optimized, single-column PDF resumes using the professional Diamond-Elite template engine.
  4. **Strategic Market Insights:** The Job Market module effectively estimates a candidate's market value, salary range, and identifies role matches on global portals like LinkedIn and Naukri.
  5. **Executive Admin Oversight:** The admin dashboard provides real-time visualization of talent analytics, submission history, and placement readiness across the user base.

- The Smart Resume Builder and Analyzer addresses the major limitations of existing systems by providing a unified, secure, and deeply analytical solution for professional advancement. Unlike traditional platforms that require sensitive data uploads or provide generic feedback, this system focuses on Local Intelligence and Factor-Based Customization, which significantly improves the precision of a candidate's professional narrative.

CHAPTER 10
CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The Smart Resume Builder and Analyzer project successfully demonstrates the development of a high-performance, privacy-centric platform that empowers professionals to own their career narrative through integrated AI auditing and elite document construction. By consolidating 10-factor resume auditing, professional PDF generation, resume-based interview prep, and market insights into a single unified "Career Cockpit," the system addresses the critical fragmentation found in current recruitment tools.

By focusing on local intelligence and deterministic scoring, the platform provides a more rigorous and secure approach to career development compared to traditional cloud-based systems. The successful integration of Llama 3.2 via Ollama ensures that users receive expert-level insights while maintaining absolute sovereignty over their sensitive professional data.

Overall, the Smart Resume Builder and Analyzer bridges the gap between raw professional experience and the sophisticated requirements of the modern job market. The system is scalable, robust, and designed with a premium aesthetic, making it an essential tool for both freshers and experienced professionals aiming for top-tier placement success.

10.2 FUTURE ENHANCEMENTS

Although the current platform provides a powerful foundation for career optimization, several strategic enhancements can be implemented to further expand its functionality. Future development could focus on Predictive AI Career Roadmapping to suggest long-term learning paths based on emerging industry trends, alongside Multimodal Mock Interview Simulation featuring real-time voice and sentiment analysis to improve candidate confidence. Furthermore, the integration of Live Market APIs would allow for more precise, real-time salary tracking and automated job application updates.

APPENDIX – A

SOURCE CODE

```
# ==========================
#    IMPORTS & SETUP
# ==========================
import streamlit as st
import pandas as pd
import base64, random, time, datetime, os, io, re, json

# PDF Processing
from pdfminer3.layout import LAParams
from pdfminer3.pdfpage import PDFPage
from pdfminer3.pdfinterp import PDFResourceManager, PDFPageInterpreter
from pdfminer3.converter import TextConverter

# UI & Media
from PIL import Image
from streamlit_tags import st_tags

# Database
import pymysql

# Visualization
import plotly.express as px

# PDF Generation
from fpdf import FPDF

# Local AI Model
import ollama

# ==========================
#    PAGE CONFIG
# ==========================
st.set_page_config(page_title="Smart Resume AI", page_icon="✨", layout="wide")

# ==========================
#    DATABASE SETUP
# ==========================
connection = pymysql.connect(
    host='localhost',
    user='root',
    password='1972'
)
cursor = connection.cursor()

cursor.execute("CREATE DATABASE IF NOT EXISTS resume_ai;")
```

```
cursor.execute("USE resume_ai;")

cursor.execute("""
CREATE TABLE IF NOT EXISTS user_data (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(255),
   email VARCHAR(255),
   score VARCHAR(10),
   role VARCHAR(255),
   timestamp VARCHAR(50)
);
""")

def insert_data(name, email, score, role):
   timestamp = str(datetime.datetime.now())
   sql = """
   INSERT INTO user_data (name, email, score, role, timestamp)
   VALUES (%s, %s, %s, %s, %s)
   """
   cursor.execute(sql, (name, email, score, role, timestamp))
   connection.commit()

# ==========================
#    PDF TEXT EXTRACTION
# ==========================
def pdf_reader(file):
   resource_manager = PDFResourceManager()
   fake_file_handle = io.StringIO()

   converter = TextConverter(resource_manager, fake_file_handle, laparams=LAParams())
   page_interpreter = PDFPageInterpreter(resource_manager, converter)

   with open(file, 'rb') as fh:
      for page in PDFPage.get_pages(fh):
         page_interpreter.process_page(page)

   text = fake_file_handle.getvalue()

   converter.close()
   fake_file_handle.close()
   return text

# ==========================
#    AI ANALYSIS (LLAMA3)
# ==========================
def analyze_resume(resume_text):
   prompt = f"""
   Analyze this resume and return JSON:
   {{
      "ATS Score": "",
```

```
        "Skills": [],
        "Weakness": [],
        "Recommended Role": "",
        "Suggestions": []
    }}

    Resume:
    {resume_text}
    """

    response = ollama.generate(
        model="llama3",
        prompt=prompt,
        options={"temperature": 0.2}
    )

    return response['response']

def clean_json(response):
    try:
        return json.loads(response)
    except:
        return None

# ==========================
#    PDF GENERATOR
# ==========================
def generate_resume_pdf(data):
    class ResumePDF(FPDF):
        def header(self):
            self.set_fill_color(30, 41, 59)
            self.rect(0, 0, 210, 12, 'F')

        def footer(self):
            self.set_y(-15)
            self.set_font("Arial", "I", 8)
            self.cell(0, 10, f"{data.get('name')} | Page {self.page_no()}", align="C")

    pdf = ResumePDF()
    pdf.add_page()
    pdf.set_auto_page_break(auto=True, margin=15)

    pdf.set_font("Arial", "B", 22)
    pdf.cell(0, 10, data.get("name", ""), ln=True)

    pdf.set_font("Arial", "", 11)
    pdf.cell(0, 8, f"{data.get('email')} | {data.get('phone')}", ln=True)

    pdf.ln(5)
```

```
    def section(title, content):
        pdf.set_font("Arial", "B", 14)
        pdf.cell(0, 8, title, ln=True)
        pdf.set_font("Arial", "", 11)
        pdf.multi_cell(0, 6, content)
        pdf.ln(2)

    section("Summary", data.get("summary", ""))
    section("Skills", data.get("skills", ""))
    section("Experience", data.get("experience", ""))
    section("Education", data.get("education", ""))

    return bytes(pdf.output())

# =========================
#     DASHBOARD CHART
# =========================
def show_dashboard():
    cursor.execute("SELECT name, score FROM user_data")
    data = cursor.fetchall()

    if data:
        df = pd.DataFrame(data, columns=["Name", "Score"])
        fig = px.bar(df, x="Name", y="Score", title="User ATS Scores")
        st.plotly_chart(fig)

# =========================
#     MAIN APP
# =========================
def run():
    st.title("    Smart Resume Builder & Analyzer")

    menu = ["Home", "Resume Analyzer", "Resume Builder", "Dashboard"]
    choice = st.sidebar.selectbox("Navigation", menu)

    # =========================
    #     HOME
    # =========================
    if choice == "Home":
        st.markdown("## AI Powered Resume System")
        st.write("""
        ✔ Analyze Resume with AI
        ✔ Get ATS Score
        ✔ Generate Professional Resume
        ✔ Store & Track Data
        """)

    # =========================
    #     ANALYZER
    # =========================
```

```
elif choice == "Resume Analyzer":
    st.subheader("Upload Resume")

    uploaded_file = st.file_uploader("Upload PDF", type=["pdf"])

    if uploaded_file:
        with open("temp_resume.pdf", "wb") as f:
            f.write(uploaded_file.getbuffer())

        text = pdf_reader("temp_resume.pdf")

        if st.button("Analyze"):
            with st.spinner("Analyzing..."):
                result = analyze_resume(text)

            parsed = clean_json(result)

            if parsed:
                score = parsed.get("ATS Score", "0")
                role = parsed.get("Recommended Role", "N/A")

                st.success(f"ATS Score: {score}")
                st.write(parsed)

                name = st.text_input("Your Name")
                email = st.text_input("Email")

                if st.button("Save Result"):
                    insert_data(name, email, score, role)
                    st.success("Saved to Database")

            else:
                st.error("AI Output Error")

# ========================
#    BUILDER
# ========================
elif choice == "Resume Builder":
    st.subheader("Create Resume")

    name = st.text_input("Name")
    email = st.text_input("Email")
    phone = st.text_input("Phone")

    summary = st.text_area("Summary")
    skills = st_tags(label="Skills", text="Enter skills")
    experience = st.text_area("Experience")
    education = st.text_area("Education")

    if st.button("Generate Resume"):
```

```
        data = {
            "name": name,
            "email": email,
            "phone": phone,
            "summary": summary,
            "skills": ", ".join(skills),
            "experience": experience,
            "education": education
        }

        pdf = generate_resume_pdf(data)

        st.download_button(
            "Download Resume",
            pdf,
            file_name="resume.pdf"
        )

    # ==========================
    #    DASHBOARD
    # ==========================
    elif choice == "Dashboard":
        st.subheader("Analytics Dashboard")
        show_dashboard()

# ==========================
#    RUN APP
# ==========================
if __name__ == "__main__":
    run()
```

APPENDIX – B

SCREENSHOTS

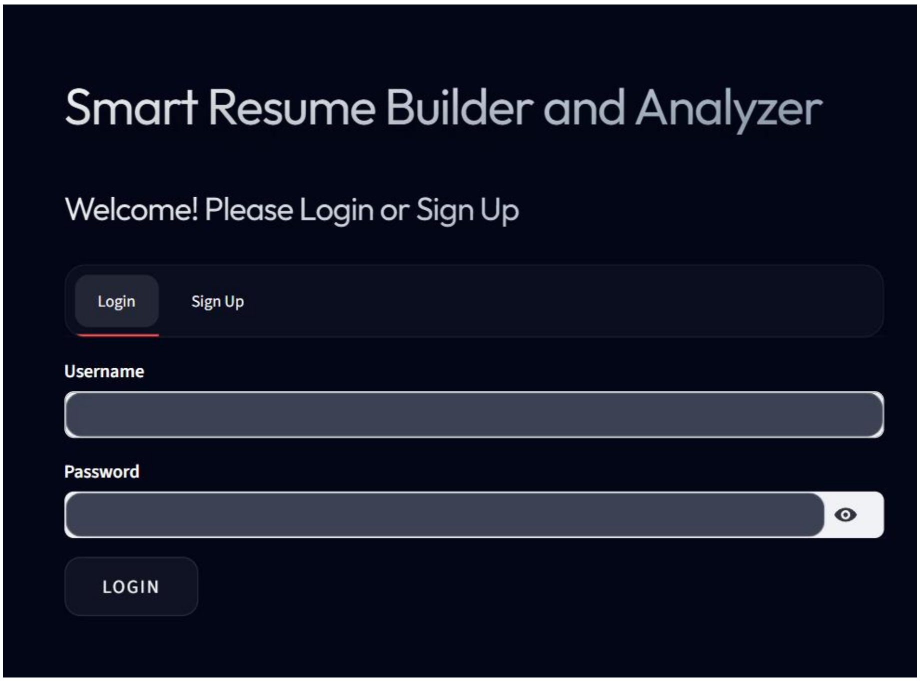

Figure B.1 Login Page

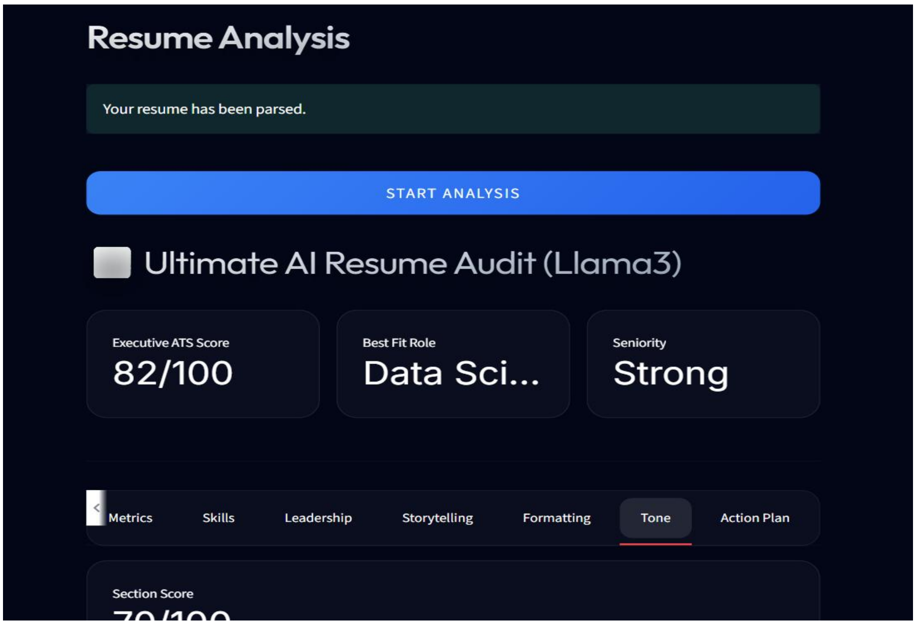

Figure B.2 Resume Analyzer

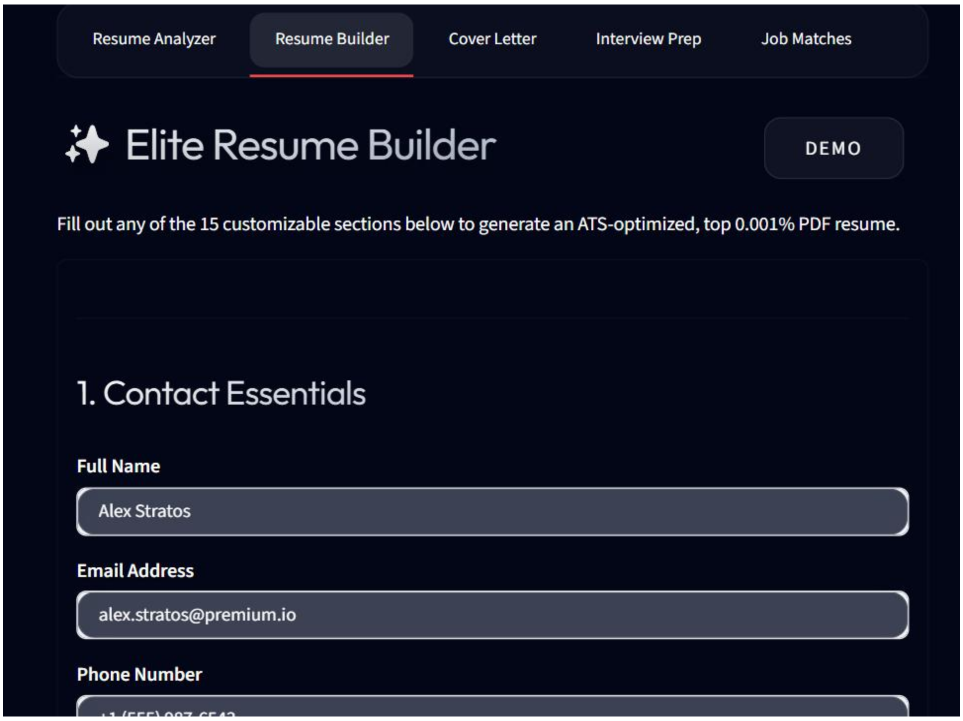

Figure B.3 Resume Builder

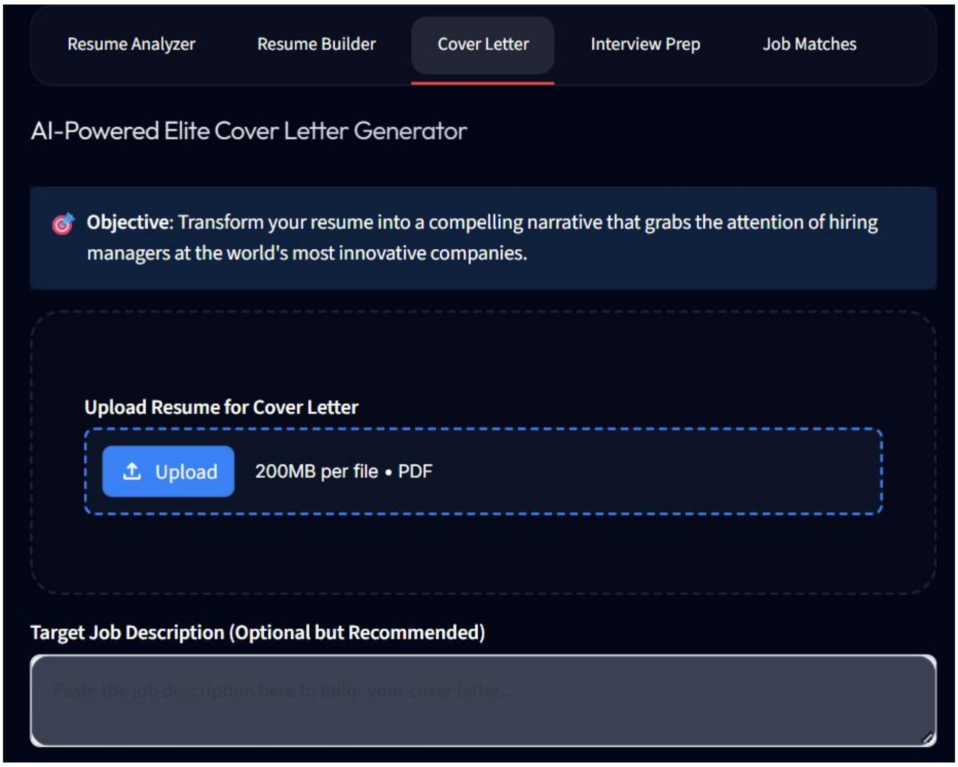

Figure B.4 Cover Letter Generator

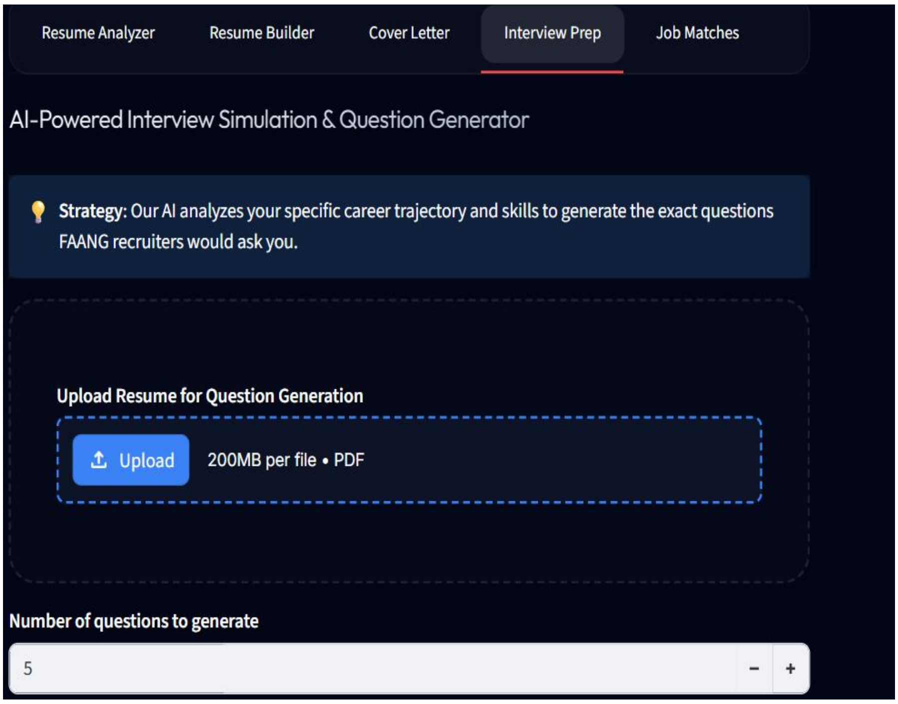

Figure B.5 Interview Question Generator

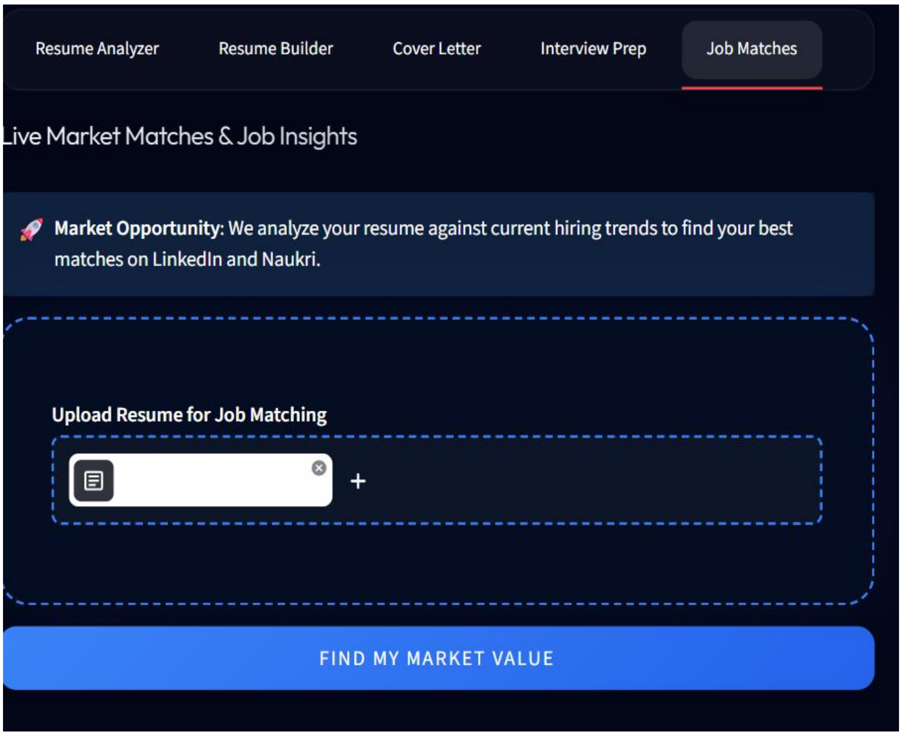

Figure B.6 Job Matching

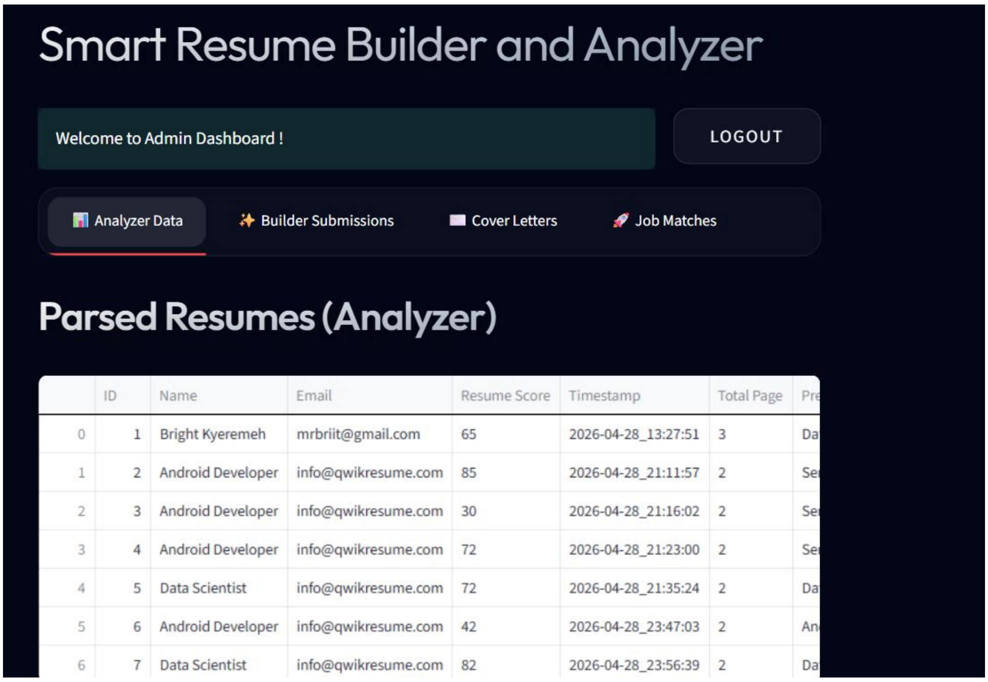

Figure B.7 Admin Dashboard

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